

Lab 18 Mini-Project: Pertussis and the CMI-PB Project

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1. Investigating Pertussis Cases by Year

Q1. Assign the CDC pertussis case number data to a data frame called `cdc` and use `ggplot` to make a plot of cases numbers over time.

We load the CDC data from the downloaded CSV file, clean the commas from the case numbers using the required adjustments from the instructor announcement, and plot the cases over time.

```
#| message: false #| warning: false library(ggplot2)
```

Load the data

```
cdc <- read.csv("pertussis_data.csv")
```

Clean the commas using the sub instructor's exact code

```
cdcNumber.of.Reported.Pertussis.Cases <- as.numeric(gsub(",", "", cdcNumber.of.Reported.Pertussis.Cases))
```

Q1 Plot: Cases over time

```
ggplot(cdc, aes(x = Year, y = Number.of.Reported.Pertussis.Cases)) + geom_point() +  
geom_line() + labs(title = "Pertussis Cases by Year (1922-2025)", y = "Number of Cases", x =  
"Year") + theme_bw()
```